

SAJEEVAN VEERIAH

Mechatronics, Robotics, Automation and AI/ML Engineer

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PROFESSIONAL SUMMARY

Multidisciplinary engineer spanning robotics, control systems, industrial automation, embedded systems, electronics, CAD, manufacturing, validation, data and applied AI/ML. Strong practical base across ROS 2, SLAM, A* planning, Kalman/EKF concepts, sensor fusion, embedded C/C++, Python, ESP32, CAN, LoRaWAN, MQTT, InfluxDB, Grafana, Siemens TIA Portal, HMI, SCADA, MES, FAT, SAT and commissioning support. Experienced in regulated manufacturing automation, automotive software validation, emissions and compliance testing, IoT deployments, production operations and engineering QA.

CORE ENGINEERING CAPABILITIES

Systems integration	Sensors, actuators, drives, embedded devices, control logic, HMI/SCADA/MES, telemetry, dashboards and operational workflows.
Robotics and autonomy	ROS 2, Gazebo, RViz, SLAM, A* path planning, Kalman/EKF state estimation, LiDAR/IMU concepts, computer vision, PID and motion control.
Automation and controls	PLC fundamentals, Siemens TIA Portal, WinCC, iFIX, HMI, SCADA, MES, Modbus, Profinet, industrial Ethernet, FAT, SAT and commissioning support.
Embedded and electronics	ESP32, STM32 concepts, embedded C/C++, Python, FreeRTOS concepts, UART, I2C, SPI, CAN, ADC, PWM, KiCad, Altium and signal conditioning.
Validation and QA	Test planning, data acquisition, CAN analysis, regression testing, defect reporting, ADR/EURO compliance, ITPs, MDRs, QA records and traceability.
Mechanical and project delivery	SolidWorks, Fusion 360, mechanical integration, 3D printing, drawings, control schematics, FMEA, risk assessment, technical documentation and stakeholder coordination.

PROFESSIONAL EXPERIENCE

Automation Engineer | JAG Process Solutions Pty Ltd | Brunswick, VIC | Jan 2026 to Jun 2026

- Worked on automation and smart factory systems for pharmaceutical, biotech, food and process clients where reliability, traceability and validation discipline were central requirements.
- Supported control logic, HMIs, process visualisation, MES, batch execution and production reporting data across automated plants, skids and package units.
- Contributed to requirements review, functional design, FAT, SAT, commissioning, qualification, troubleshooting, diagnostics, handover readiness and technical documentation.
- Integrated field devices, sensors, drives and process equipment with HMI, control logic, production data and operator workflows while considering safety and maintainability.

Product Development Test Engineer | Ford Motor Company via Invenio | Little River, VIC | Oct 2025 to Jan 2026

- Worked in Feature Software Integration and Validation on vehicle software integration and ADAS validation for T6 Ranger and Everest programmes.
- Executed feature vehicle, breadboard and regression testing for readiness, running changes, OTA updates and sign-off evidence across vehicle variants.
- Used Vector CANoe, CANalyzer and diagnostic tools to capture and analyse CAN bus data for fault isolation, verification and evidence-based defect reporting in JIRA.

Technical Assistant | ABMARC | South Geelong, VIC | Jul 2024 to Aug 2025

- Supported vehicle emissions and compliance testing against Australian Design Rules and EURO standards across multiple vehicle platforms.
- Operated emissions test equipment, instrumentation and data acquisition systems with focus on repeatable, auditable and technically defensible results.
- Prepared compliance, QA and regulatory documentation for certification, audits and reporting while analysing test data to identify deviations.

Consultant Engineer - Internet of Things | DuxTel Pty Ltd | Greater Geelong, VIC | May 2024 to Aug 2024

- Designed and deployed end-to-end IoT systems linking sensors, embedded devices, gateways, cloud platforms and data pipelines.
- Built real-time ingestion, monitoring and visualisation using MQTT, LoRaWAN, InfluxDB and Grafana, with documentation and handover support.

PROFESSIONAL EXPERIENCE CONTINUED

IoT Projects Administrator | DuxTel Pty Ltd | Greater Geelong, VIC | Feb 2024 to May 2024

- Configured and integrated IoT devices into ChirpStack-based LoRaWAN networks for low-power, wide-area deployments.
- Developed an ESP32 GPS tracking and environmental sensing system with firmware configuration, device provisioning, testing and deployment workflows.

Production Worker and Cellar Hand | IDL | Moorabool, VIC | Aug 2022 to Feb 2024

- Ran and monitored food and beverage production equipment, maintained throughput and quality, completed checks and reported faults in a scheduled production environment.
- Supported hygiene, traceability, sampling, record-keeping and quality monitoring across regulated production and cellar operations.

Production Line Operator | Carbon Revolution | Waurin Ponds, VIC | Sep 2021 to Dec 2021

- Set up and operated manufacturing equipment for high-performance carbon fibre wheels while maintaining quality, safety, maintenance and first-level troubleshooting requirements.

Undergraduate Engineer | Thornton Engineering Australia Pty Ltd | Corio, VIC | Mar 2018 to Mar 2020

- Supported engineering and QA for structural steel fabrication delivered to Australian Standards and client specifications.
- Prepared Inspection Test Plans, Manufacturer Data Reports and quality records; reviewed drawings for inspection planning, material traceability and compliance.

SELECTED ENGINEERING PROJECTS

- Autonomous Navigation and ROS 2 Robotics: Built robotics software concepts using ROS 2, Gazebo, RViz, A* path planning, SLAM, Kalman/EKF state estimation, odometry, IMU fusion, simulation and modular control nodes.
- ESP32 Clinical Ataxia Assessment Concept: Developed an embedded measurement concept using ESP32, IMU, ToF, Hall and magnetometer sensing, firmware logic, data acquisition and analysis workflows.
- IoT GPS and Environmental Monitoring Platform: Developed ESP32 GPS and environmental sensing with LoRaWAN/MQTT telemetry, gateway/device configuration, InfluxDB/Grafana dashboards and deployment documentation.
- Smart Factory and Process Visualisation: Worked across HMI/process visualisation, MES and batch-system concepts where operator usability, traceability, diagnostics and maintainability must be engineered together.

ENGINEERING PRACTICE AND DELIVERY

- Project delivery: Requirements interpretation, work planning, technical documentation, cross-functional coordination, stakeholder updates, issue escalation and handover readiness.
- Testing and validation: Structured test plans, data capture, verification evidence, regression testing, FAT/SAT support, commissioning support, defect tracking and root cause analysis.
- Manufacturing and quality: QA records, traceability, inspection planning, FMEA exposure, safety procedures, production troubleshooting, process discipline and continuous improvement.
- Hands-on integration: Sensors, actuators, embedded devices, wiring, CAN data, IoT gateways, instrumentation, CAD/mechanical integration, prototype concepts and production systems.

EDUCATION, MEMBERSHIPS, TRAINING AND LANGUAGES

Bachelor of Mechatronics Engineering (Honours) , Deakin University, VIC	Mar 2018 to Jul 2025
Higher National Diploma, Mechatronics, Robotics and Automation Engineering , Cardiff Metropolitan University	2014 to 2016
Professional membership: Member, Engineers Australia	
Licence: Current Victorian Driver Licence	
Training: Lean Six Sigma Foundation; JIRA and Agile; KAIZEN; LinkedIn Learning Industrial Automation and IoT Foundations	
Languages: English, Tamil and Sinhala	

TECHNICAL KEYWORD MATRIX

Robotics and control	ROS 2, Gazebo, RViz, SLAM, A* planning, EKF, sensor fusion, LiDAR, IMU, OpenCV, PID, motion control, autonomous systems.
Automation and industrial systems	PLC fundamentals, Siemens TIA Portal, WinCC, iFIX, HMI, SCADA, MES, batch execution, Modbus, Profinet, FAT, SAT, commissioning support.
Embedded and software	ESP32, STM32 concepts, embedded C/C++, Python, FreeRTOS concepts, UART, I2C, SPI, CAN, MQTT, LoRaWAN, Linux, Git, REST APIs.
Electrical, CAD and hardware	CAN bus, CAN FD, Vector CANoe, CANalyzer, instrumentation, data acquisition, KiCad, Altium, SolidWorks, Fusion 360, schematics, mechanical integration.
Quality and delivery	Test planning, verification, validation, FMEA, QA records, ITPs, MDRs, ADR, EURO, ISO, IEC, AS/NZS exposure, technical writing, JIRA and Agile.